3/05/26

GFG_160_DSA

Day-001

P1 : Second largest Element

Problem statement : Given array arr, return the second largest distinct element from an array. If the second largest element doesn't exist return -1.

largest   2nd largest

12	35	1	10	34	1

→ 34.

Sample Test cases :

Input :

12	35	1	10	34	1

Output : 34

Input :

10	5	10

Output : 5

constraints

$2 \le arr.size \le 10^5$

$1 \le arr[i] \le 10^5$

Input :

10	10	10

Output : -1

Approach 1: (using sorting)

→

12	35	1	10	34	1	35

after sorting :

1	1	10	12	34	35	35

→ return the unique 2nd last element

complexity analysis :

Time complexity : $O(n\log n)$ $(n\log n + n)$

Space complexity : $O(1)$

Approach 2 : In two iterations

12	35	1	10	34	1	35

→ find max in 1st iteration

→ find max value (less than the previous) in 2nd iteration.

complexity analysis:

T.C : $O(2n) \sim O(n)$

S.C : $O(1)$

Approach 3 : In one iteration.

→ look for max & keep track of the previous updated value & 2nd max value.

→ | 12 | 35 | 1 | 10 | 34 | 1 | 35 |
|---|---|---|---|---|---|---|
| ↑ | ↑ | ↑ | ↑ | ↑ | ↑ | ↑ |

sec	max
12	~~-∞~~
34	~~12~~
	35

complexity analysis:

T.C. ⇒ $O(n)$

S.C → $O(1)$